

projection of $\vec{u}$ on $\vec{v}$
 $\Rightarrow \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|} \cdot \frac{\vec{v}}{\|\vec{v}\|}$

Topic: Change of Basis

Question 1. Find the standard matrix $[\mathbf{R}]$ for the rotation $\mathbf{R}$ in $\mathbb{R}^3$ about the positive x -axis by an angle θ . Note: you can assume that $\mathbf{R}$ is a linear map!

Question 2. Let $\mathbf{T}$ be the reflection on $\mathbb{R}^2$ about the line ℓ which passes through the origin and makes an angle θ with positive x -axis in counter-clockwise direction. Find the standard matrix $[\mathbf{T}]$. Hence use it to determine $\mathbf{T}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$.

Question 3. The linear transformation $\mathbf{T} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a composition of the counter-clockwise rotation by $\pi/6$ about the origin and followed by the projection on the line $y = \sqrt{3}x$. Determine the standard matrix $[\mathbf{T}]$.

Question 4. Consider the linear transformation $\mathbf{T} : \mathcal{M}_{2 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 2}(\mathbb{R})$ defined by

$$\mathbf{T}\left(\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\right) = \begin{bmatrix} a+b+2c+5f & a+2c-d+7e \\ 2a+b+7c+2d-e+f & 2b-c+d \\ 2a+b+7c+2d & 3a-7c+10d \end{bmatrix}.$$

Is $\mathbf{T}$ an isomorphism? Justify your answer. (You may use **SymPy** or some online-software for the computation involved.)

Question 5. Let $\mathbf{S}$ and $\mathbf{T}$ be linear operators on a finite dimensional space $\mathcal{V}$ such that $\mathbf{TS}$ is invertible.

- Prove that $\mathbf{S}$ is invertible, and deduce that $\mathbf{T}$ and $\mathbf{ST}$ are invertible.
- Show that the statement in part (a) is false if $\mathcal{V}$ is an infinite-dimensional vector space. You may want to consider $\mathcal{V} = \mathcal{P}(\mathbb{R})$ and the following operators $\mathbf{S}$ and $\mathbf{T}$. Here, $\mathbf{S}(f) = xf$ while $\mathbf{T}(f) = (f - f(0))/x$. If $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, then $\mathbf{T}(f) = a_1 + a_2x + \cdots + a_nx^{n-1}$.

Question 6. For each of the following pairs of bases $\mathcal{B}$ and $\mathcal{A}$ for a vector space $\mathcal{V}$, find the change-of-basis matrix $\mathbf{P}$ that changes $\mathcal{B}$ -coordinates into $\mathcal{A}$ -coordinates. That is, determine the matrix $[\mathbf{I}]_{\mathcal{B}}^{\mathcal{A}}$.

- $V = \mathbb{R}^2$, $\mathcal{A} = \{(1, 0), (0, 1)\}$, and $\mathcal{B} = \{(a_1, a_2), (b_1, b_2)\}$, where a_1, a_2, b_1, b_2 are some fixed scalars in $\mathbb{R}$ and $a_1b_2 - a_2b_1 \neq 0$.
- $V = \mathcal{P}_2(\mathbb{R})$, $\mathcal{A} = \{2x^2 - x, 3x^2 + 1, x^2\}$ and $\mathcal{B} = \{1, x, x^2\}$.

Question 7. Let $\mathbf{T}$ be the linear operator on $\mathcal{P}_1(\mathbb{R})$ defined by $\mathbf{T}(p(x)) = p'(x)$, the derivative of $p(x)$. Let $\mathcal{B} = \{1, x\}$ and $\mathcal{A} = \{1 + x, 1 - x\}$. Find $[\mathbf{T}]_{\mathcal{A}}^{\mathcal{A}}$.

Question 8. Consider the following matrix $\mathbf{A}$ and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$\mathbf{A} = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

- (a) Find the change-of-basis matrix from $\mathcal{A}$ to the standard basis $\mathcal{B}$ for $\mathbb{R}^3$.
- (b) Let $\mathbf{T}$ be the matrix-multiplication corresponding to $\mathbf{A}$. Determine $[\mathbf{T}]_{\mathcal{A}}^{\mathcal{A}}$.